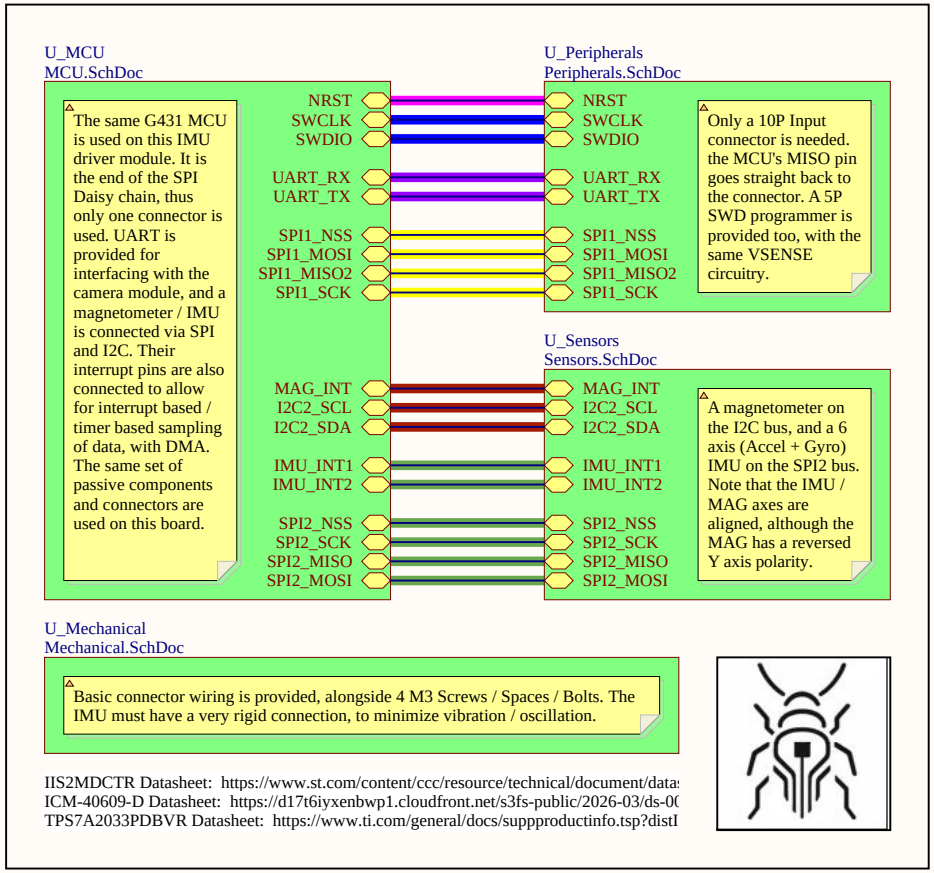
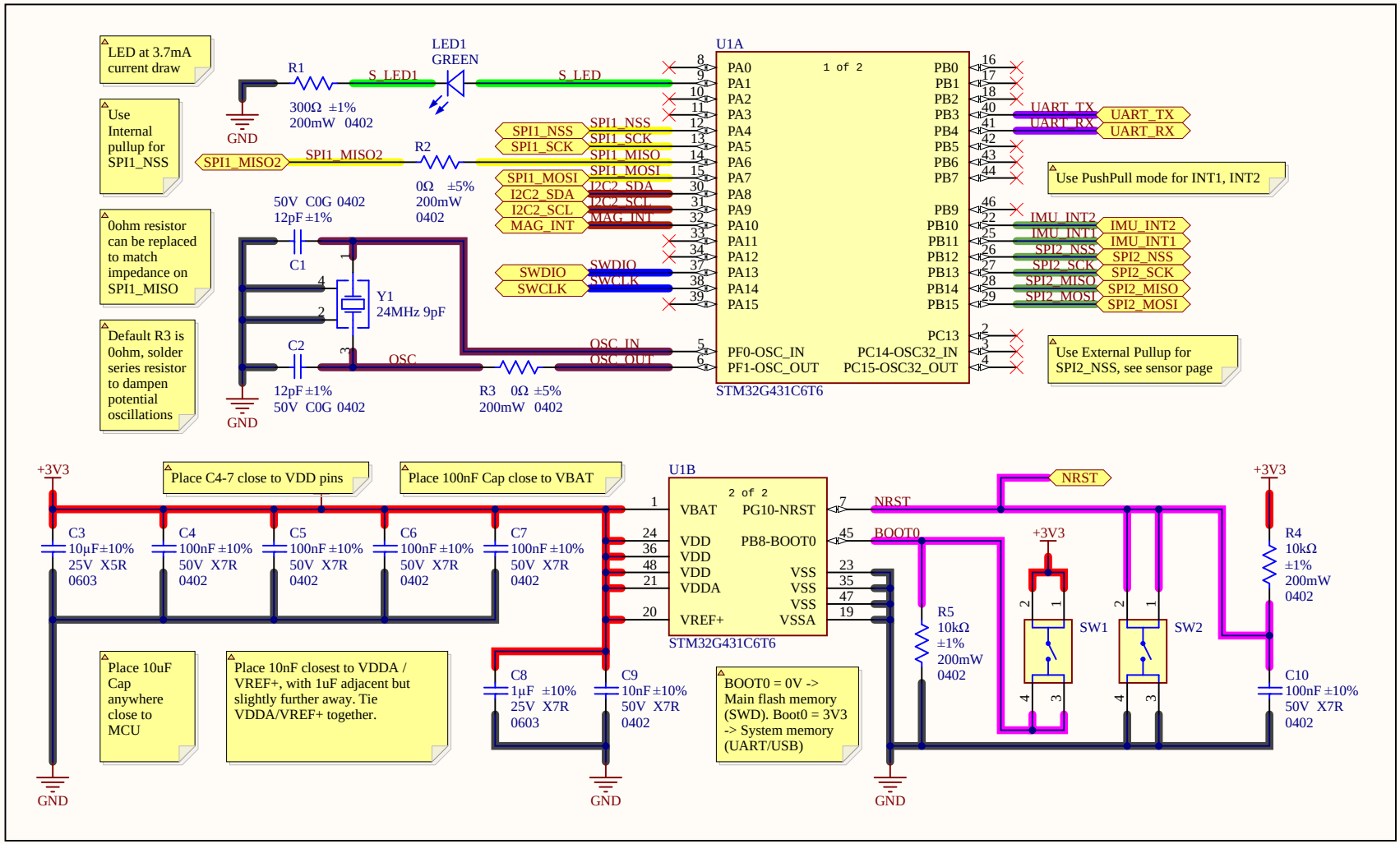
UQ Biorobot New Tracker
IMU Driver Module

System Overview - All Sheets



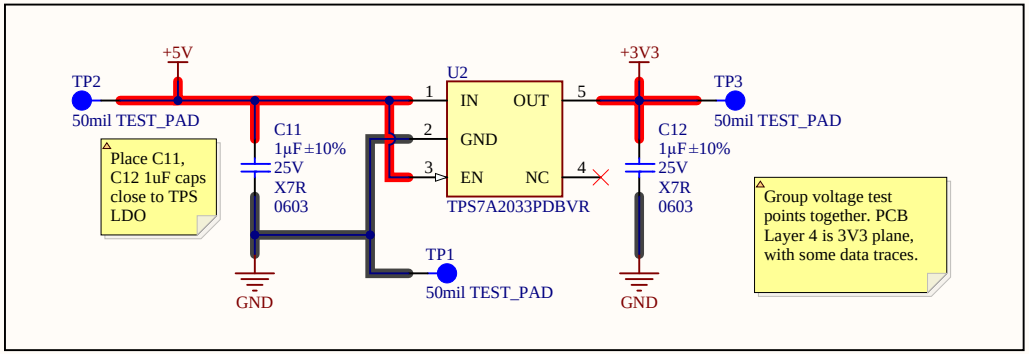
Title: Overview		
Size: A4	Number: 1	Revision: 1
Date: 7/05/2026		Sheet 1 of 5
File: C:\Users\..\Overview.SchDoc		Drawn By: Xuyang Liu

STM32G431C6T6 Microcontroller

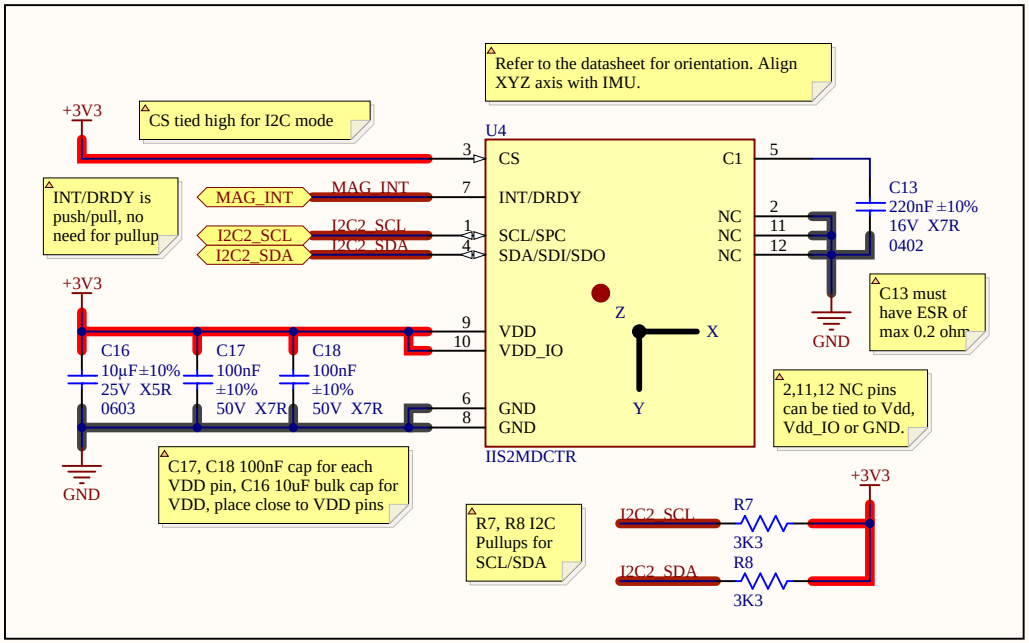


Title: MCU		
Size: A4	Number: 2	Revision: 1
Date: 7/05/2026		Sheet 2 of 5
File: C:\Users\..\MCU.SchDoc		Drawn By: Xuyang Liu

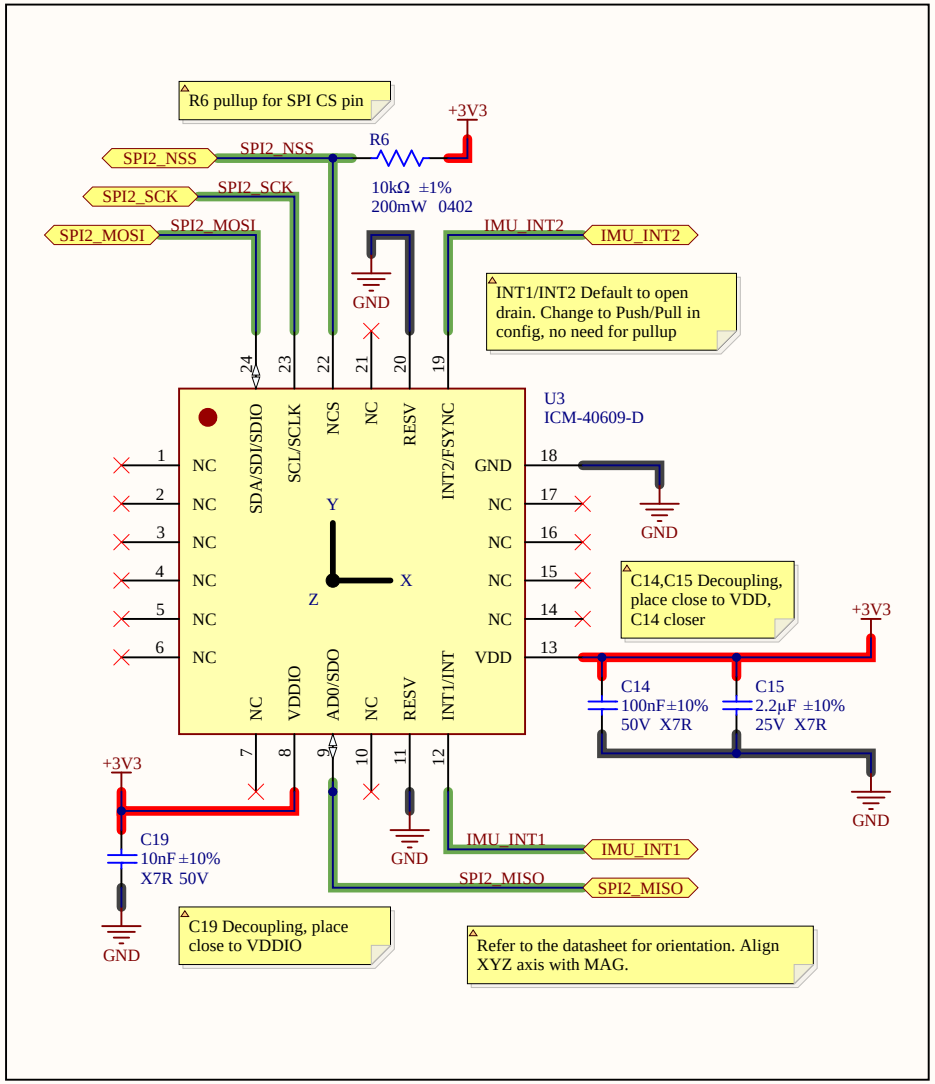
5 -> 3.3V LDO and Test Points



IIS2MDC Magnetometer (I2C)

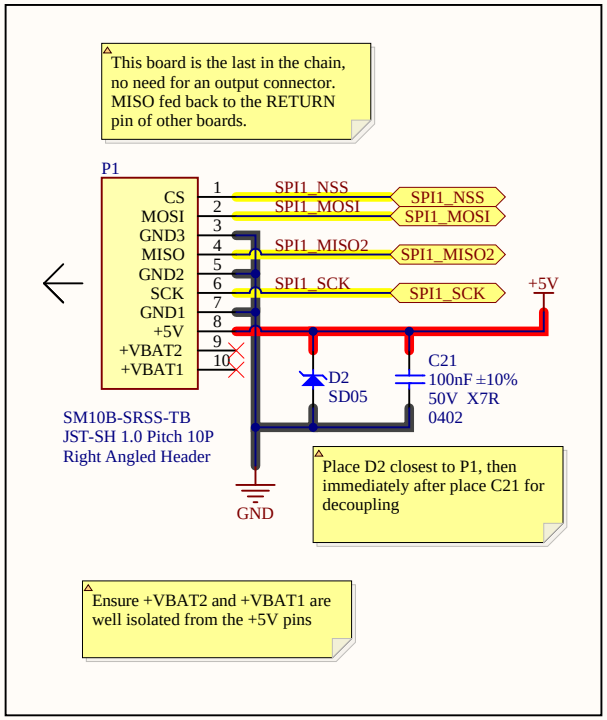


ICM-40609-D 6 axis IMU (SPI)

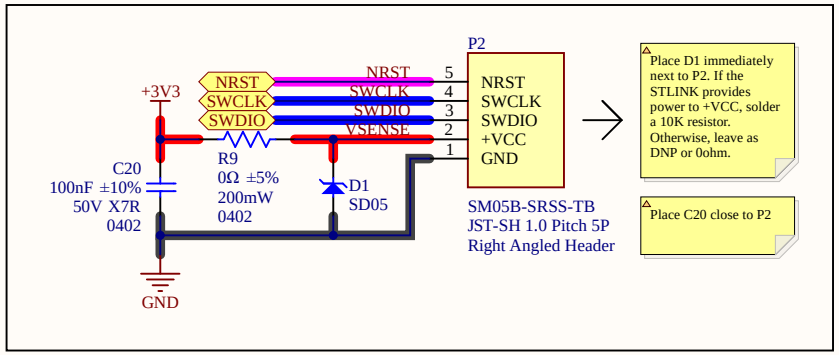


Title: Sensors		
Size: A4	Number: 3	Revision: 1
Date: 7/05/2026	Sheet 3 of 5	
File: C:\Users\..\Sensors.SchDoc	Drawn By: Xuyang Liu	

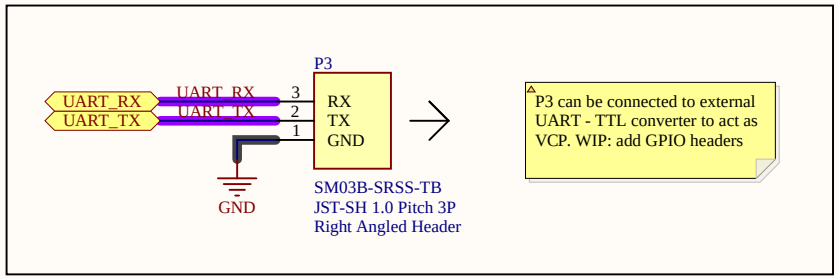
SPI Daisy Chain Power/Data Input



SWD Programming Header



UART header



Title: Peripherals		
Size: A4	Number: 4	Revision: 1
Date: 7/05/2026	Sheet 4 of 5	
File: C:\Users\..\Peripherals.SchDoc	Drawn By: Xuyang Liu	

M3 Nuts

HW2
Nut, M3.0

HW5
Nut, M3.0

HW8
Nut, M3.0

HW11
Nut, M3.0

M3 Plated Holes

H1
M3 Mounting Holes

H2
M3 Mounting Holes

H3
M3 Mounting Holes

H4
M3 Mounting Holes

M3 Screws

HW3
Screw, M3.0 x 0.6

HW6
Screw, M3.0 x 0.6

HW9
Screw, M3.0 x 0.6

HW12
Screw, M3.0 x 0.6

M3 Spacers

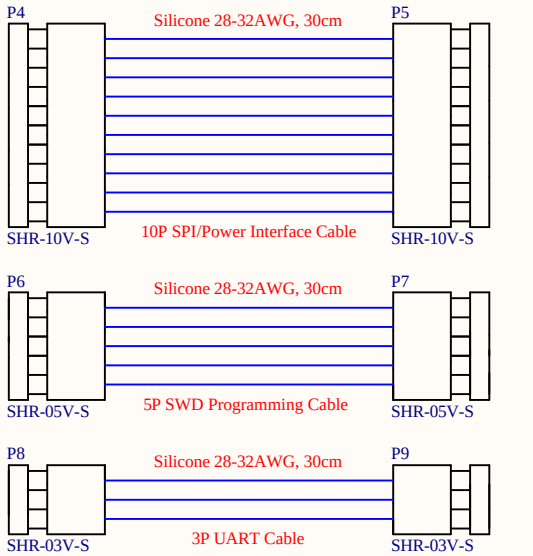
HW1
M3.0 x 05

HW4
M3.0 x 05

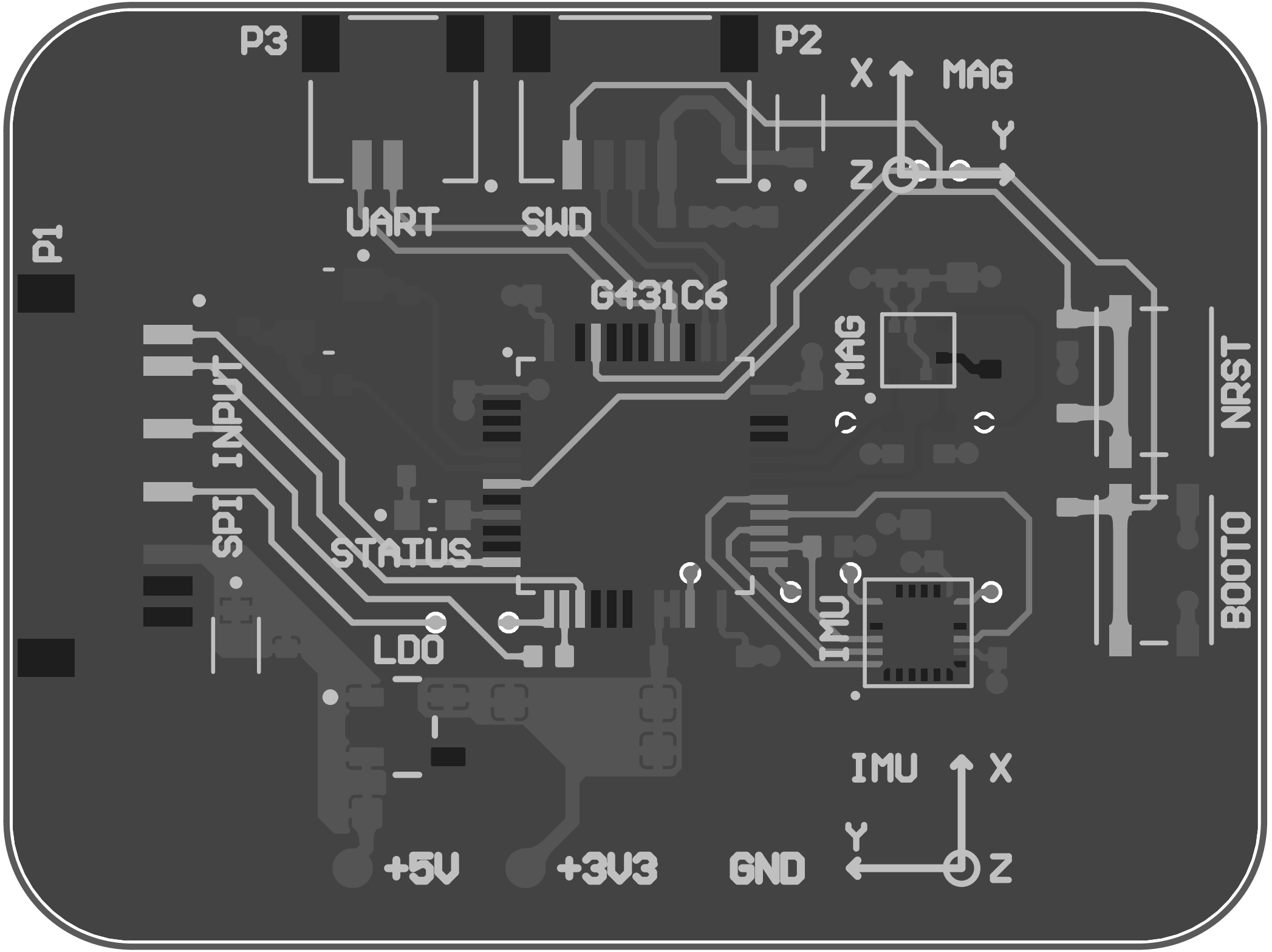
HW7
M3.0 x 05

HW10
M3.0 x 05

JST-SH Connector Housing (3P, 5P, 10P)



Title: Mechanical		
Size: A4	Number: 5	Revision: 1
Date: 7/05/2026		Sheet 5 of 5
File: C:\Users\..\Mechanical.SchDoc		Drawn By: Xuyang Liu


P3
P2
P1
UART
SWD
G431C6
SPI INPUT
STATUS
LDO
MAG
IMU
NRST
BOOT0
+5V
+3V3
GND
X
MAG
Y
Z
IMU
X
Y
Z